

MES Remedial Questions-

1. A fire alarm activates when smoke concentration exceeds a threshold. Identify the trigger mechanism used (event or time) in embedded system and explain its importance.
2. A company is designing an aircraft autopilot system. Why is an embedded system preferred over a general-purpose computer?
3. Analyze the suitability of embedded systems in railway signal control systems using four technical parameters.
4. RISC Vs CISC comparison

1. Role of the TMOD and TCON registers in 8051 timers.
2. **What are the contents of registers A, R0, R7 and memory locations 30H and 31H after the following code runs: -**

```
MOV A, #5
MOV R7, #40H
MOV R0, #30H
MOV 31H, #14H
MOV @R0, A
INC R0
MOV R7, @R0
```

3. **What are the contents of registers and memory locations**

```
ORG 0000H
MOV A, #30H
MOV 60H, #40H
CJNE A, 60H, NOTEQ
MOV A, #00H
SJMP DONE
NOTEQ: MOV A, #0FFH
DONE: END
```

4. Write a 8051 program to find average of three 8-bit numbers stored at 45H, A1 H, and 32H. Store the final result at 71H. Also draw flow chart .
5. Write an 8051 program to find the largest of three 8-bit numbers stored at 40H, 41H, and 42H. Store the result at 50H. Draw the flowchart.

1. Show the interfacing diagram of 8051 with ADC, also write ALP.